

4.2 Points on the Coordinate Plane – Part 1

Lesson Introduction

 Benchmark	MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x - or y -axis as the line of reflection when two ordered pairs have an opposite x - and y -coordinate.		 Learning Targets	- I can plot integers using different scales. - I can plot rational numbers in the form of decimal and fractional ordered pairs. - I can select an appropriate scale given a set of ordered pairs.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.GR.4.1 Identify the origin and axes in the coordinate system. Plot and label ordered pairs in the first quadrant of the coordinate plane.		N/A	
 Essential Questions	- How can you determine the scale for ordered pairs that include decimals? - How can you determine the scale for ordered pairs that include fractions? - How can you use the least common multiple to determine the scale for ordered pairs that include fractions?			
 Lesson Narrative	In Lesson 1, students plotted points with integer values on one-unit scales. This lesson builds on what students learned previously to include plotting integer coordinates on a set of axes using different scales. Students will practice choosing a correct scale with decimal units and fractional units and work collaboratively to plot fraction and decimal numbers on the coordinate plane.			
 Component Summary	4.2.1 Warm-Up (~5 min.)	Career profile of an interior designer (<i>SE p. 138</i>)	4.2.4 Collaboration (10 min.)	Plot decimals on a coordinate plane (<i>SE p. 141</i>)
	4.2.2 Refresher (5 min.)	A reminder of common, but foundational, key terms. (<i>SE p. 139</i>)	4.2.5 Collaboration (10 min.)	Determine best scale to use and plot rational numbers (<i>SE p. 142</i>)
	4.2.3 Guided Instruction (10-15 min.)	Plot points on a fractional scale (<i>SE p. 139</i>)	4.2.6 Try it (5 min.)	Create a scale to plot decimal points and plot points on a fractional scale independently (<i>SE p. 144</i>)
	4.2.7 Wrap-Up (~5 min.)	Reflect on understanding of learning targets (<i>SE p. 145</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Coordinate plane - Axes - Coordinate - Origin - Quadrant - Scale - Least Common Multiple Additional Terms: N/A		(Optional) Dry erase boards (Optional) Dry erase markers - Graph Paper	4.2.1 Warm-Up contains a video to accompany the question prompts.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle with estimating the location of rational ordered pairs. - Students may not understand how to find the least common denominator. - Students may struggle with linking the scale to the least common denominator.	- Help students create appropriate graphs and scales by asking for the largest and smallest x and y values that need to fit in the allotted space. - Students are still grasping rational numbers as well as negative numbers. Consider reviewing what these numbers look like in a visual context to help understanding. - If time is an issue, consider assigning 4.2.6 Try It for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 4.2.3 Guided Instruction provides an opportunity for students to engage in a Think-Pair-Share routine to formalize their explanations with a partner before engaging with the written-response questions.	MA.K12.MTR.4.1: Discussion and Reflection This lesson encourages students to work collaboratively with their partners to reason about plotting points on a coordinate plane. Encourage students to analyze their partner's explanations and critique or build upon responses as necessary. Through partner work, students engage in mathematical reasoning and have the opportunity to justify and explain their work.
 Differentiate Instruction	Highlight and make space for the differentiated approaches to sensemaking, which are embedded throughout this lesson. - For verbal processors, or students who learn best when thinking and reasoning with others, the 4.2.4 Collaboration will drive their learning and offer opportunity for making meaning of fluency with scales. - For students who learn best with trial and error or seeing the work in action, consider providing a graphing utility as they work through the 4.2.4 Collaboration.	
Lesson Notes:		

4.2.1 Warm-Up: Interior Designer

Component Overview: In this Warm-Up, students will learn about a career in interior design. After watching a video showing Soledad Alzaga discussing her career, students will consider what career they are interested in and what skills they need for that career.



4.2.1 Warm-Up: Interior Designer

After getting a degree in graphic design, Soledad Alzaga built her career and company as an interior designer. Interior designers make interior spaces functional, safe, and beautiful by determining space requirements and selecting decorative items such as colors, lighting, and materials. Beyond being creative and having good taste, Soledad says to own a successful interior design business, it is necessary to have the ability to work and communicate with others, be organized, have business knowledge, and to have math and measurement skills.

1. What is something that interests you that you would like to spend your career doing?

Answers will vary. I am interested in becoming a Youtuber/Influencer.

2. What skills do you think are needed to be successful in that career?

Answers will vary. I need to have marketing skills and skills for making videos.



Do you know someone who is an interior designer or works in graphic design? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.

4.2.2 Refresher: Key Terms With the Coordinate Plane

Component Overview: Students review key vocabulary from Lesson 1 and Grade 5 related to the coordinate plane.

4.2.2 Refresher: Key Terms with the Coordinate Plane

Several key terms related to the coordinate plane were used throughout Lesson 1. Below, you can find definitions for each of these key terms.



The **coordinate plane** is a plane determined by two perpendicular number lines, called axes. The axes intersect at the origin. Each point in the coordinate plane is represented by a pair of coordinates that represent the direction and distance from each axis. The origin is represented by the coordinate pair, $(0, 0)$.



The **axes** of a graph are the horizontal and vertical number lines used in a coordinate plane system.



A **coordinate** is a number used to locate a point on a number line. A coordinate is one of the numbers in an ordered pair that locates a point on a coordinate plane.



The **origin** is the point of intersection of the x - and y -axes in a rectangular coordinate system, where the x -coordinate and y -coordinate are both 0.



A **quadrant** is any of the four regions separated by the axes in a coordinate plane.



Consider vocabulary strategies that align to your classroom and your students' learning needs.

- Frayer Model
- Word wall
- Anchor chart



Prompt students to engage in increasingly precise mathematical discussions using the vocabulary provided. The following might be common, informal examples of words to replace:

- Replace “center” with “origin.”
- Replace “part” with “quadrant.”
- Replace “lines” with “axes.”

4.2.3 Guided Instruction: Plotting Points With a Fractional Scale

Component Overview: In this Guided Instruction, students will practice plotting points that are fractions on a coordinate plane. Students will need to create a fractional scale to accurately plot each point. Students will also draw and create their own graph using fractional points. This activity is also available via Desmos.



4.2.3 Guided Instruction: Plotting Points with a Fractional Scale

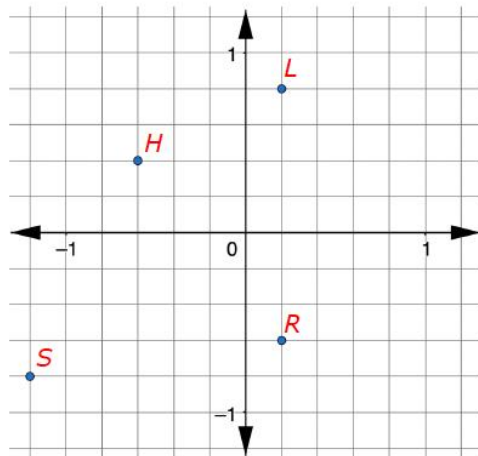
1. Four points are shown.

Point L	Point H	Point S	Point R
$(\frac{1}{5}, \frac{4}{5})$	$(-\frac{3}{5}, \frac{2}{5})$	$(-1\frac{1}{5}, -\frac{4}{5})$	$(\frac{1}{5}, -\frac{3}{5})$

- a. Based on the denominators of each fraction, what scale should be used for the grid?

$\frac{1}{5}$

- b. Find and label the points with the given letters on the coordinate plane.



2. Another set of four points is shown.

Point M	Point L	Point C	Point A
$(\frac{2}{3}, \frac{1}{6})$	$(\frac{1}{2}, -\frac{1}{3})$	$(-\frac{5}{6}, -\frac{1}{2})$	$(-\frac{1}{6}, 1)$

- a. List the denominators of each fraction.

2, 3, 6



Students may struggle to plot the fractional points on the graph. Encourage students to reason about the appropriate scale by asking questions like:

- How many fractional parts should the space between 0 and 1 be broken up into, to accurately plot each point?
- Is there another fractional scale that can be used for this graph?
- Does the scale need to be the same for both the x- and y- axes?

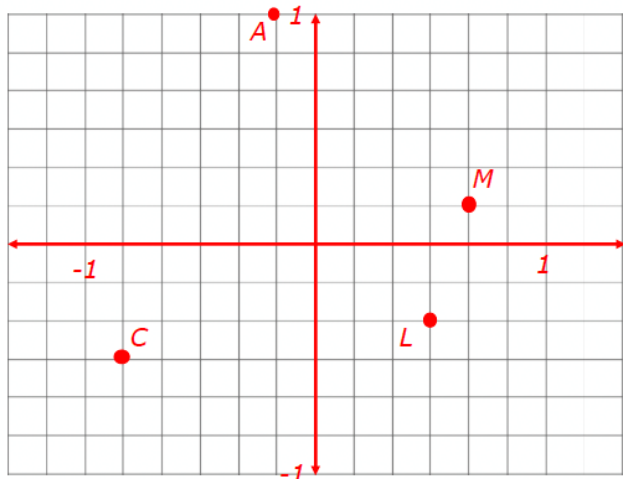
4.2.3 Guided Instruction: Plotting Points With a Fractional Scale (continued)

- b. Zoe says she does not want to label her axes by halves because she will only be able to approximate the location of the thirds. She also says she does not want to label by thirds because there is no easy way to approximate the location of the halves. Christopher says she should label her axes by sixths.

Explain the benefit of using Christopher's plan.

Answers will vary. Christopher's plan will allow him to plot fractions with denominators of 2 and 3 because he will be able to write those as equivalent fractions with a denominator of 6.

- c. Draw and label coordinate axes with the scale you chose and plot the points.



When deciding on a scale for plotting ordered pairs with rational values, consider the least common multiple (LCM) of each of the denominators to determine the proper increments.



Think-Pair-Share

Engage students in a Think-Pair-Share routine as they discuss and complete question 2b. This will give students time to analyze each student's response and determine why Christopher's plan is beneficial. Then students discuss with a partner to encourage mathematical discourse and reasoning.

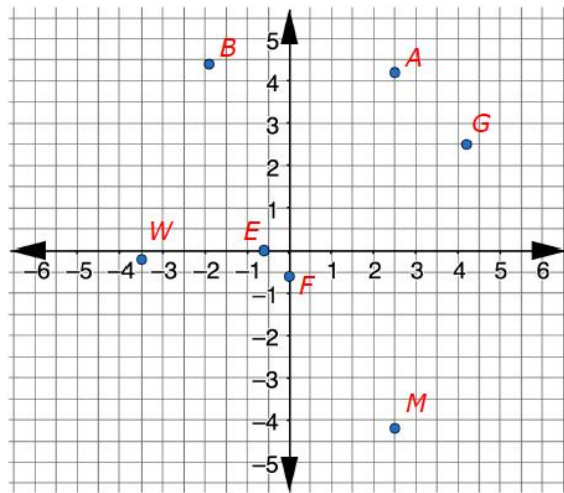
4.2.4 Collaboration: Plotting Points With a Decimal Scale

Component Overview: In this Collaboration, students will work with a partner to plot decimals on a coordinate plane. Students will practice plotting points on a graph that uses whole numbers for the scale and discuss why or why not using this type of graph may work.



4.2.4 Collaboration: Plotting Points with a Decimal Scale

1. A coordinate plane is shown.



- a. Working with your partner, describe the scale that is being used for each axis.

Answers will vary. The scale is 0.5 or $\frac{1}{2}$ because it takes 2 lines to get to each whole number. 1 whole divided by 2 is 0.5 or $\frac{1}{2}$.

- b. Locate each point on the coordinate plane and label it with the appropriate letter.

Point G (4.2, 2.5)	Point B (-1.9, 4.4)	Point F (0, -0.6)	Point W (-3.5, -0.2)
Point E (-0.6, 0)	Point A (2.5, 4.2)	Point M (2.5, -4.2)	



To encourage students to begin to draw conclusions about the scale on the graph, consider asking guiding questions like:

- When graphing Point B, should the x -coordinate be closer to -1 or -2 ?
- What do you notice about where each unit on the scale falls? How many lines apart are they? What does this tell you about the scale?
- What do you know about points that don't fall directly on the lines where the numbers are?

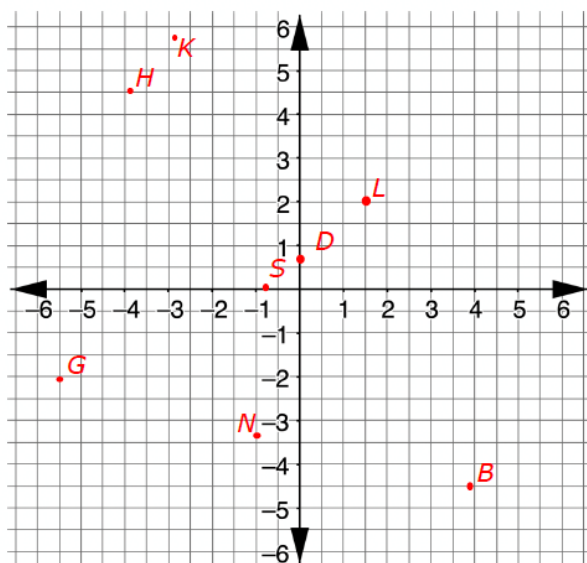
4.2.4 Collaboration: Plotting Points With a Decimal Scale (continued)

2. Working with your partner, plot and label the points using the given letters on the grid.

Point <i>L</i> (1.5, 2)	Point <i>B</i> (3.9, -4.5)	Point <i>G</i> (-5.5, -2.1)	Point <i>D</i> (0, 0.7)
Point <i>K</i> (-2.8, 5.7)	Point <i>S</i> (-0.8, 0)	Point <i>N</i> (-1, -3.4)	Point <i>H</i> (-3.9, 4.5)



Consider allowing students to graph the points using a graphing utility if they are struggling to plot the points on the graph.



- a. Which points were easy to plot?
Answers will vary. Point L, Point D, Point H and Point S
- b. Which points were more difficult to plot?
Answers will vary. Point B, Point G, Point N, Point K



Specifically have students make connections to which points were easy to plot and which were more difficult. Encourage students to reason about what scales might be appropriate for decimal numbers and fractions, making note that points that were not exactly halfway between two numbers might have been more difficult to plot.

4.2.4 Collaboration: Plotting Points With a Decimal Scale (continued)

3. On your own, answer the following question:

When plotting ordered pairs with decimal values, how is it best to determine the scale to use when setting up the axes on a grid?

My Reasoning

Answers will vary. To determine the scale to use when plotting ordered pairs with decimals determine any common factors in the coordinates. For example, if all of the x -coordinates have one decimal place and the tenths place is 0 or 5, then the scale I would choose would be .5 or $\frac{1}{2}$.

Share your reasoning with your partner and ask for feedback on any parts that were unclear or missing. Make any necessary edits to your reasoning.



Consider providing students with sentence starters to help scaffold their thinking on their own and when speaking with their partner.

- “To determine the best scale to use when plotting decimal values, you can...”
- “I know how to choose which scale to use by considering...”

4.2.5 Collaboration: Choosing the Right Scale

Component Overview: In this Collaboration, students work with a partner to determine appropriate scales for graphs based on the data presented. Students choose a graph to best represent a data set and explain why their choice is the best and why the other graphs are not as accurate.



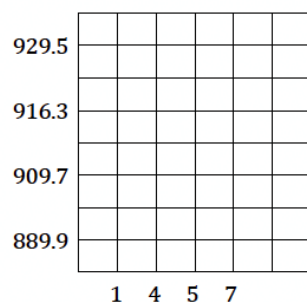
4.2.5 Collaboration: Choosing the Right Scale

When representing data, the scale of the grid is important to effectively and accurately communicate information.

- The table shows the total Florida tourism industry employees, in thousands, for years since 2002.

Years Since 2002	Total Tourism Industry Employees (in thousands)
1	889.9
4	909.7
5	916.3
7	929.5

To plot the data, Andrea drew the coordinate grid shown.



Discuss with your partner why Andrea's coordinate grid would not effectively nor accurately communicate the information about Florida tourism industry employees.

Answers will vary. The scale would not accurately communicate information because it is not a consistent scale. For example, some of the units on the x-axis represent 1 year but some of the units represent 3 years.



MA.K12.MTR.6.1: Reasonableness

Students will discuss why the given graph is not appropriate for the data with a partner. Prompt students with questions like:

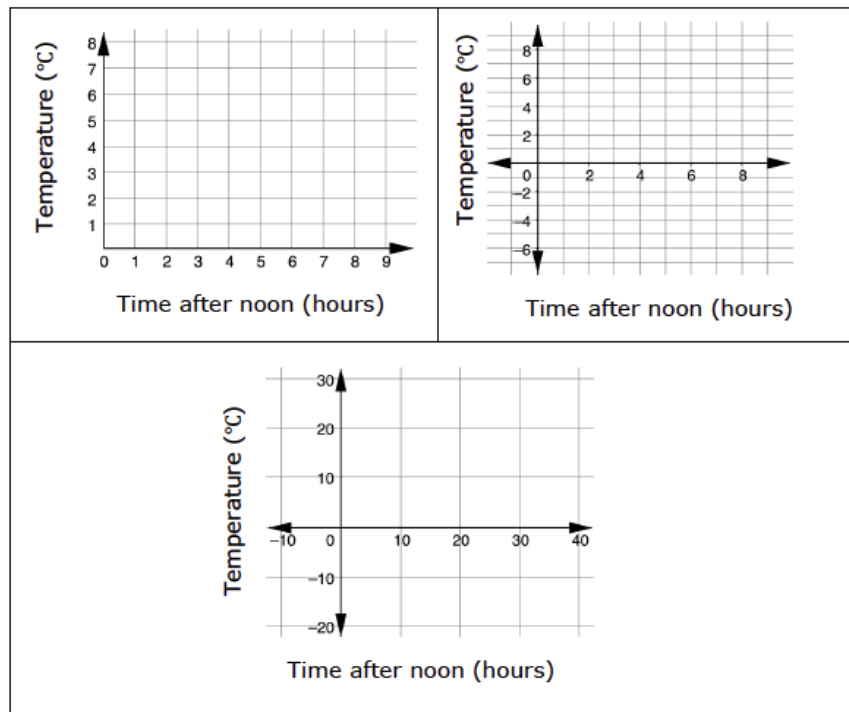
- What is the scale given on the graph?
- How would the graph look when the points are plotted?
- When graphed, would the data accurately show the growth?
- What would be a good scale to use for this set of data?

4.2.5 Collaboration: Choosing the Right Scale (continued)

2. The data shown in the table are average autumn temperatures, in degrees Celsius ($^{\circ}\text{C}$), in Carmathen, Wales.

Time After Noon (hours)	Temperature ($^{\circ}\text{C}$)
0	5
1	4
2	3
3	2
4	1
5	-2
6	-3
7	-4
8	-4

Three coordinate grids are shown, each using different scales.



Think-Pair-Share

Leverage the Think-Pair-Share instructional routine to facilitate meaningful discourse using the prompts in questions 2a and 2b below. Prompt students to individually write their responses to parts a and b before sharing with a partner. Give students the opportunity to refine their responses based on what their partner shared before eliciting responses whole group. Specifically listen for students who change their responses based on what their partner shared and have those students share to continue encouraging growth mindset.

4.2.5 Collaboration: Choosing the Right Scale (continued)

- a. Explain which grid most accurately represents the data.

Answers will vary. The grid in the upper-right corner most accurately represents the data because the x -axis uses a scale that includes all numbers between 0 through 8 while the y -axis includes all the numbers between -4 and 5, which are the ranges of values listed in the table above.

- b. Explain why the other two grids may not be as accurate.

Answers will vary. The grid in the upper left corner is not accurate because it does not include negative values for temperature. The grid at the bottom is not accurate because the scale is too large to accurately plot any of the data and to see any relationships or trends.



As students consider the coordinate grids shown, circulate and ask questions to aid in their analysis like:

- Look at each graph. Are there any points you cannot plot?
- When you graph the data, does it look different on each graph? Does this matter? Why?
- What do you notice about the scales on each of these coordinate grids? Why might that be important when plotting the data points?

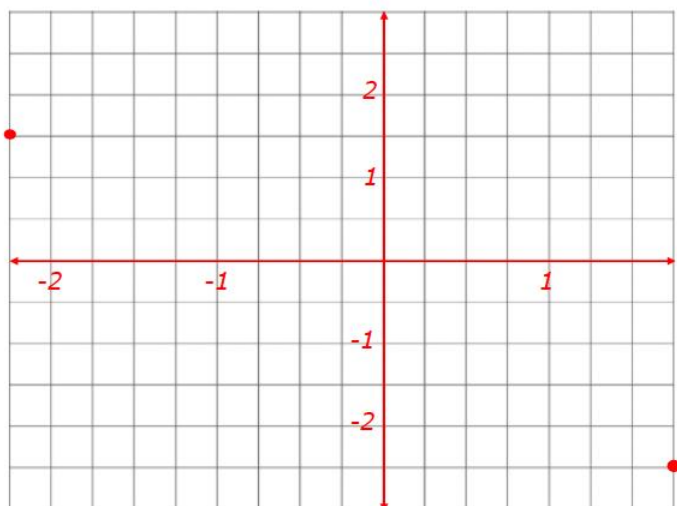
4.2.6 Try It: Plotting Points With Fractional and Decimal Scales

Component Overview: In this Try It, students will be given a blank graph and will need to create a decimal scale and plot decimal points accordingly. Then, students will plot points using a fractional scale. They will need to decide which fractional amount to use for the graphing by finding a common denominator.



4.2.6 Try It: Plotting Points with Fractional and Decimal Scales

- For the points $(1.75, -2.5)$ and $(-2.25, 1.5)$, draw and label coordinate axes with an appropriate scale, and plot the points.

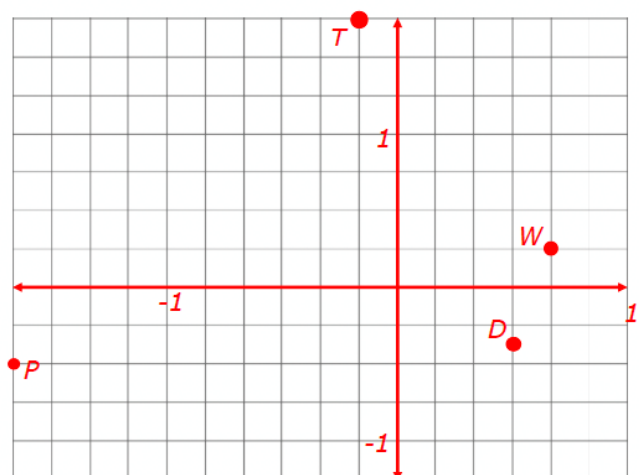


Consider allowing students to plot the points on a dry erase board before writing in their workbooks. This will allow students to use trial and error to arrive at an answer, as they can erase a whiteboard more easily.

4.2.6 Try It: Plotting Points With Fractional and Decimal Scales (continued)

2. Draw and label coordinate axes with an appropriate scale. Then, plot and label the points.

Point W	Point D	Point P	Point T
$\left(\frac{2}{3}, \frac{1}{4}\right)$	$\left(\frac{1}{2}, -\frac{1}{3}\right)$	$\left(-\frac{5}{3}, -\frac{1}{2}\right)$	$\left(-\frac{1}{6}, \frac{7}{4}\right)$



Students may struggle to determine an appropriate scale because the fractions have different denominators. Further scaffold this section through questioning like:

- What is a common denominator that can be used for all of these coordinates? What's the least common multiple, and how can you find it?
- How can you convert the fractions, so they all have the same denominator?
- Do the x - and y -axes need to have the same scale? Why or why not?

4.2.7 Wrap-Up: Reflection

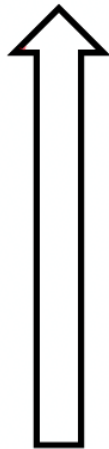
Component Overview: In this Wrap-Up, students will reflect on their understanding of the lesson's learning targets. Consider using this Wrap-Up as a guide for which students will need reteaching or reinforcing of the lessons the next day.



4.2.7 Wrap-Up: Reflection

1. Shade the arrow up to the statement that best describes your current level of understanding.

Answers will vary.



I'm so confident, I could explain plotting ordered pairs made up of fractions and/or decimals to someone else!

I can get to the right answer, but I don't understand it well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. I checked the boxes I think I do not understand.

I tried hard and I listened, but I am finding this challenging. I checked the boxes I think I do not understand.

2. Check the skills that you currently understand.

Answers will vary.

- ☐ Determining the correct scale
- ☐ Identifying the LCM of the denominators for coordinates that include fractions
- ☐ Plotting ordered pairs that contain fractions
- ☐ Plotting ordered pairs that contain decimals



Consider providing additional support for struggling readers by reading the checklist of skills out loud or showing examples of each skill in context.